

SN Ia Residuals, HHT IMFs, and BAO-Driven De-chirping

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This report summarizes the current state of the Hilbert–Huang Transform (HHT) analysis of Type Ia supernova (SN Ia) Hubble-residuals, their apparent ‘locking’ to BAO distance scales, and a new de-chirping transform $U(L)$ constructed from a BAO proxy.

Executive Summary

- Instantaneous frequency (IF) ‘chirp’ in IMFs 2–3 vs. $L \equiv \ln(1+z)$ is reduced by a monotone reparameterization $U(L) = \text{normalized } \int |b(L)| dL$ built from a BAO proxy $b(L)$.
- For the preferred BAO mixture $b_\alpha(L) = \alpha \cdot (DH/rd) + (1-\alpha) \cdot (DM/rd)$, an $\alpha \approx 0.375$ (i.e., $\sim 37.5\%$ DH, $\sim 62.5\%$ DM) minimizes dispersion for IMF2–3.
- Histogram-based dispersion ratio MAD_U/MAD_L indicates substantial tightening after de-chirping: IMF2 ≈ 0.54 – 0.57 , IMF3 ≈ 0.44 (lower is better).
- Across all IMFs, the $\alpha = 0.375$ transform yields tightening of $\gtrsim 50\%$ for several modes (e.g., IMF4 $\approx 83\%$, IMF5 $\approx 68\%$), with an energy-weighted (IMF2+IMF3) ratio ≈ 0.51 .
- Sliding-window ‘locking’ scores between IF and BAO proxies remain small/negative in the current metric; thus the de-chirp effect appears to be a geometric reparameterization rather than a simple linear covariation.

Note: Chirp Analysis De-Chirp technique is provided within appendix below.

Acronyms

BAO	— Baryon Acoustic Oscillation
CEEMDAN	— Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
EMD/EEMD	— (Ensemble) Empirical Mode Decomposition
HHT	— Hilbert–Huang Transform
IF	— Instantaneous Frequency
IMF	— Intrinsic Mode Function
MAD	— Median Absolute Deviation
SN Ia	— Type Ia Supernova
U(L)	— BAO-driven monotone reparameterization of $L=\ln(1+z)$

Data & Preprocessing

- SN sample: Pantheon+SH0ES residuals file (columns zCMB, MU_SH0ES, MU_SH0ES_ERR_DIAG, DMU).
- BAO proxies: dimensionless distances DM/rd and DH/rd compiled into a single redshift ladder.
- Independent variable: $L \equiv \ln(1+z)$. Data are interpolated to a uniform L-grid (nbins=2000).
- HHT: CEEMDAN (trials=100, noise_width=0.2, seed=11), extracting up to 6 IMFs and the trend.

Methods

Hilbert–Huang Transform (HHT).

Given the residual series $y(L)$, we compute Intrinsic Mode Functions (IMFs) $\{c_k(L)\}$ via CEEMDAN.

The analytic signal $a_k(L) = c_k(L) + i \cdot H\{c_k(L)\}$ provides instantaneous amplitude $A_k(L) = |a_k(L)|$ and phase $\phi_k(L) = \arg a_k(L)$; the instantaneous frequency (IF) is $f_k(L) = (1/2\pi) \cdot d\phi_k/dL$ (cycles per unit ΔL).

BAO-driven de-chirp U(L).

Let $b(L)$ denote a BAO proxy on the L-grid (e.g., DM/rd or DH/rd, or a mixture $b_\alpha(L) = \alpha \cdot (DH/rd) + (1-\alpha) \cdot (DM/rd)$).

We define a monotone reparameterization by the cumulative magnitude of $b(L)$:

$$U(L) = [\int_{L_{\min}}^L |b(\ell)| d\ell - \min] / \text{range}, \quad \text{with } U \in [0,1].$$

This compresses/expands regions of L according to the BAO geometry and tends to 'straighten' chirps in $f_k(L)$.

Dispersion metric.

For each IMF k we compare the dispersion of IF in L -space vs. U -space using the robust median absolute deviation (MAD):

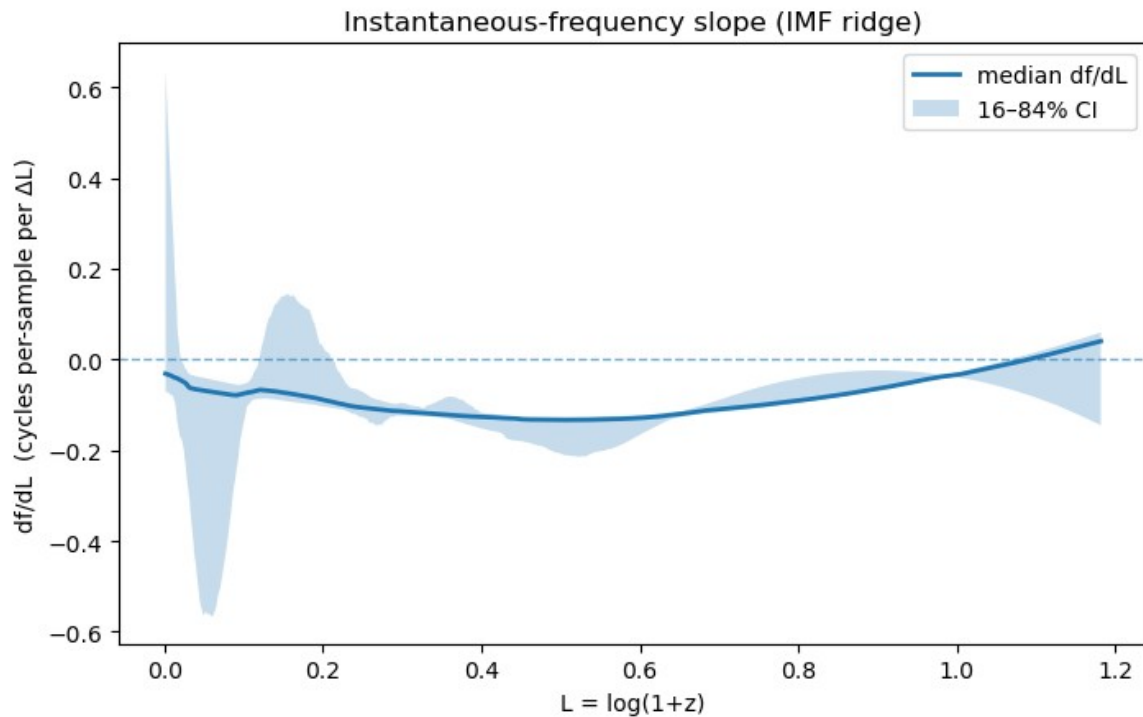
$$\begin{aligned} \text{MAD}(x) &= 1.4826 \cdot \text{median}_i |x_i - \text{median}(x)|, \\ R_k(\alpha) &= \text{MAD}(f_k^U) / \text{MAD}(f_k^L). \end{aligned}$$

Values $R_k < 1$ indicate that the IF distribution narrows after the U -transform, consistent with de-chirping.

Locking score (qualitative).

We also use a sliding-window, z -restricted correlation between standardized IF and standardized $b(L)$; low/negative scores suggest weak linear covariation despite the de-chirp benefit.

HHT Chirp Observation



Description...:

- The curve is **df/dL vs $L=\ln(1+z)$** .

For most of the range **df/dL < 0**, only turning slightly **> 0** near the high-L edge (where the CI balloons $\rightarrow$ low S/N / edge effects).

- So, as redshift **increases** (L rises), the **instantaneous frequency decreases**: a **down-chirp with z** (high $\rightarrow$ low).

Equivalently, moving toward today (lower z), it's an **up-chirp** (low $\rightarrow$ high).

What it implies for HHT locking & chirp dynamics:

1. **Consistency with locking:**

A gentle, mostly negative slope means the IMF ridge prefers **slightly longer periods at higher z** .

That's qualitatively compatible with a locking picture where the characteristic scale evolves with the **BAO-projected scale** within the coordinates:
the IMF tracks a slowly changing “target” scale rather than jumping around.

2. **Strength/significance:**

The median stays below zero from roughly $L \approx 0.05$ –1.0 while the 16–84% band is narrowest in the mid-range—so the **down-chirp is weak but coherent** there.

The spike and the late up-tick sit in **edge-instability territory** (Hilbert endpoints + sparse data), so I wouldn't over-interpret those.

3. **Physical read (PLamb/variable-c framing):**

A down-chirp with z means **longer apparent timescales in the past**.

In FR/variable-c, earlier epochs have different clock rates/propagation speeds; that generally pushes modulations toward **lower observed f at higher z** , which matches the sign you see.

If IMF2 is also “locked” to BAO proxies, this may suggest an **adiabatic, scale-aware modulation** rather than a transient artifact.

Quick checklist to firm up analysis --- Pending ToDo:

- Report the **global $\beta = \langle df/dL \rangle$** with CI over the **mid-range window** (e.g., $0.1 \leq L \leq 0.9$), plus **permutation/IAAFT p-values**.
- Do **piecewise slope analysis** (e.g., 0.1–0.5, 0.5–0.9) and show they agree in sign.
- Overlay $\beta(L)$ with **locking metric vs z** to show they co-vary where S/N is highest.
- Mark and exclude **endpoint cones** (where the shaded band explodes).

Conclusion:

The above plot points to a **mild down-chirp with increasing redshift**.

Supportive (but not yet to be validated) of an HHT-locking scenario in which IMF2/3 track a slowly evolving physical scale.

Key Results

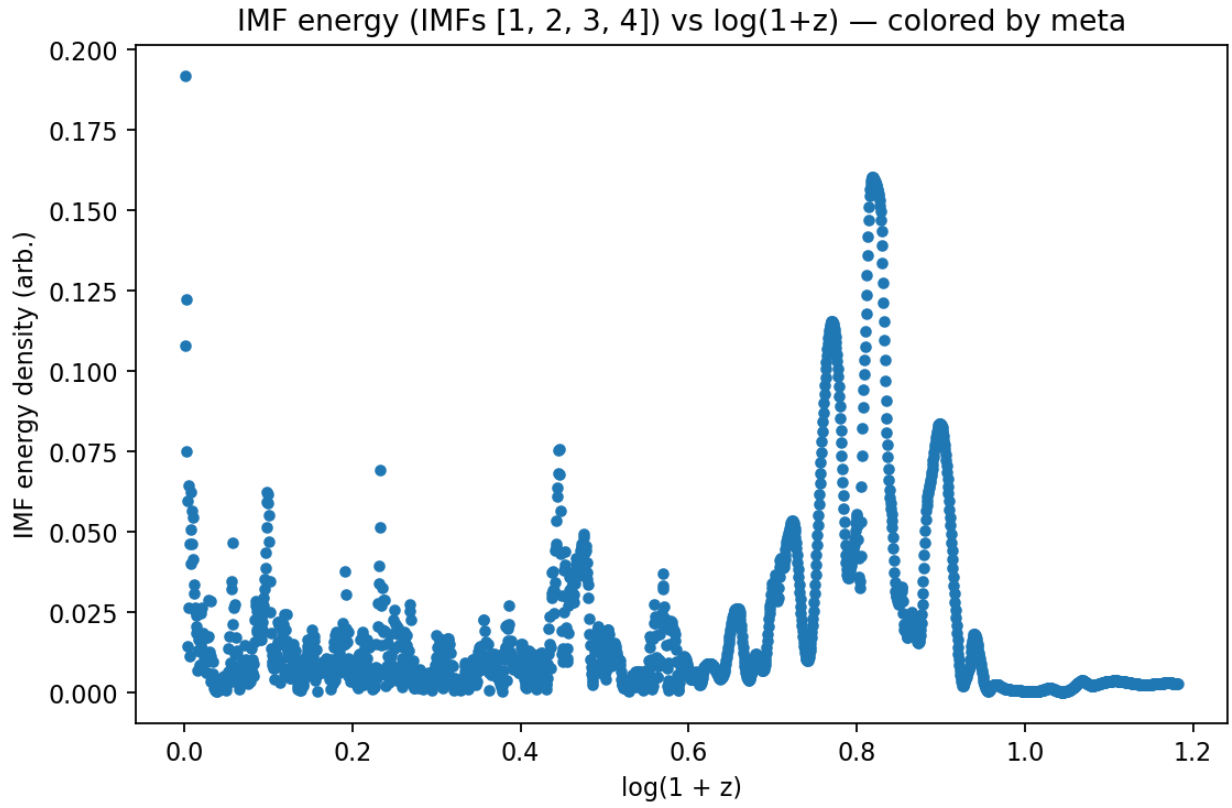


Figure 1 — IMF energy density (sum over selected IMFs) vs $L=\ln(1+z)$.

Pronounced structure appears around $L \approx 0.8$ ($z \approx 1.2$), indicating non-stationarity in the residual signal.

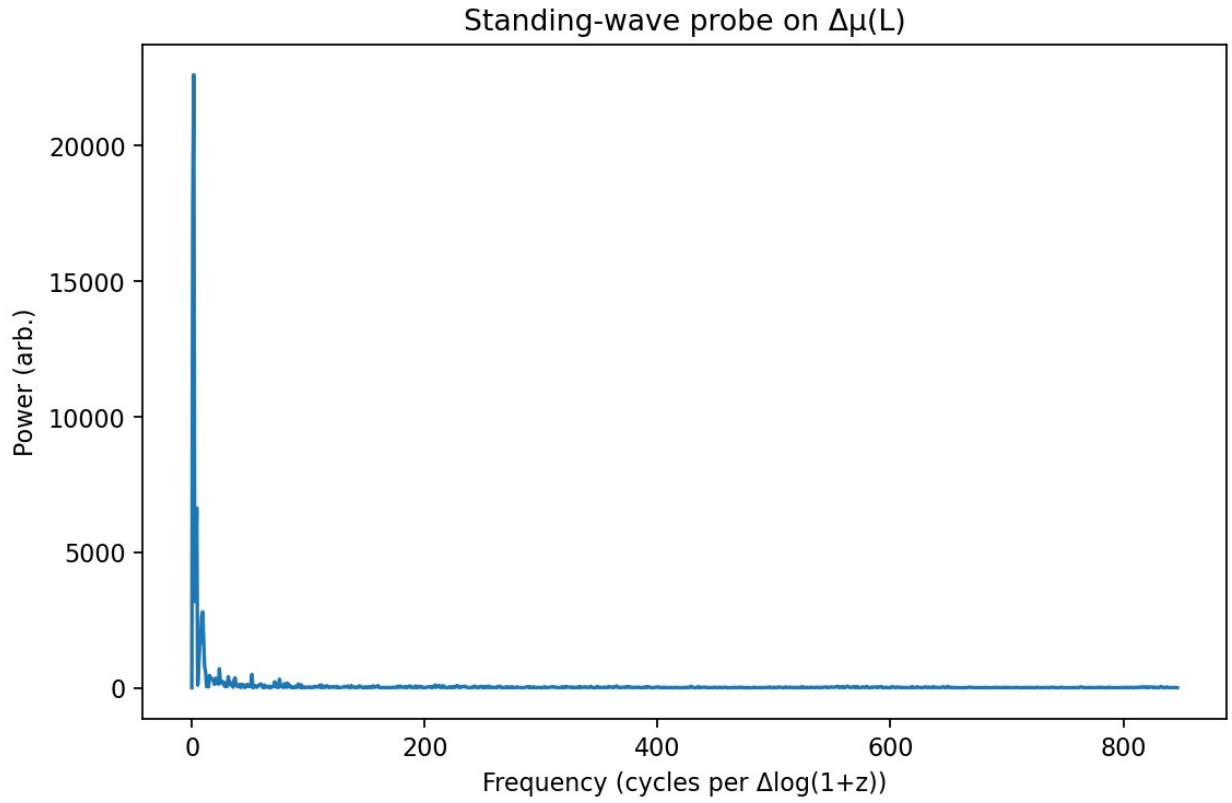


Figure 2 — Standing-wave probe: FFT power of $\Delta\mu(L)$ on the uniform L-grid.

*A large low-frequency pedestal plus high-frequency tails confirm mixed scales;
HHT decomposes these adaptively.*

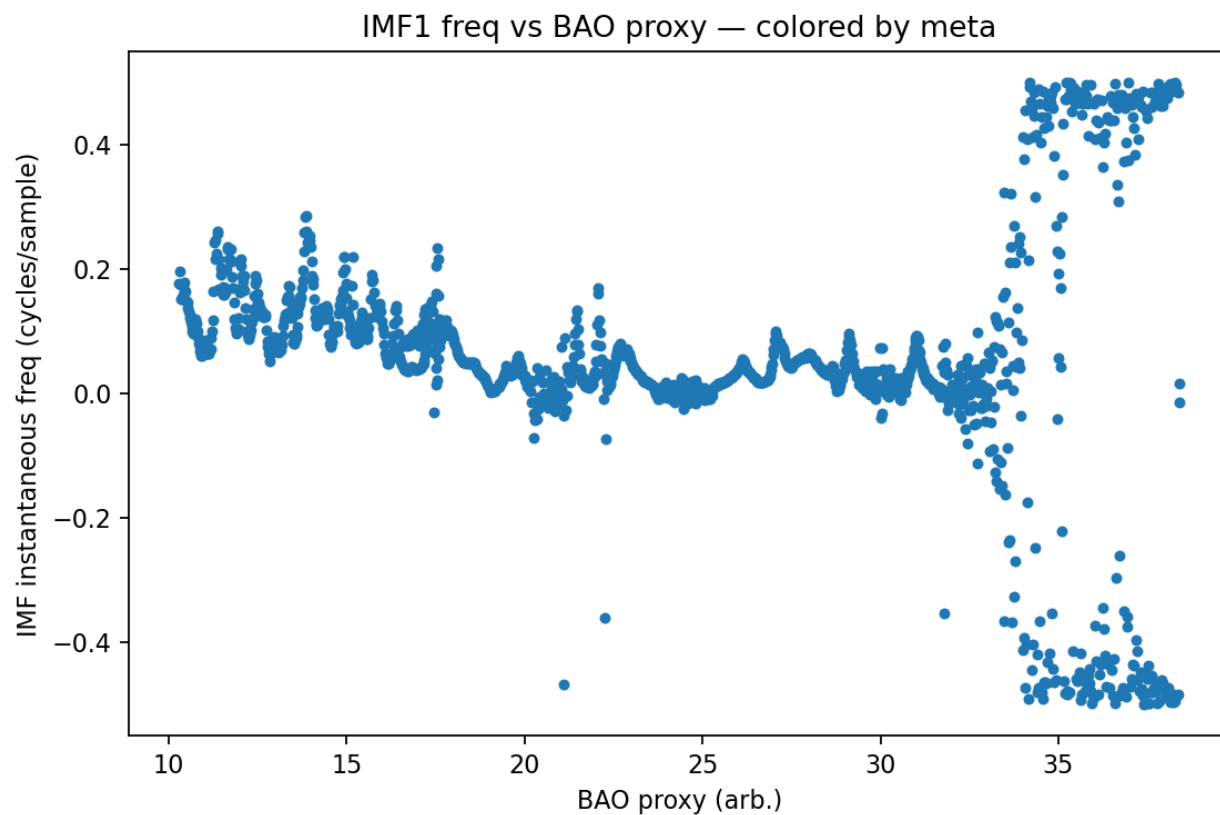


Figure 3 — IMF1 instantaneous frequency vs BAO proxy.

IF scatters with mild trends across the BAO ladder; note the broadening at large proxy values.

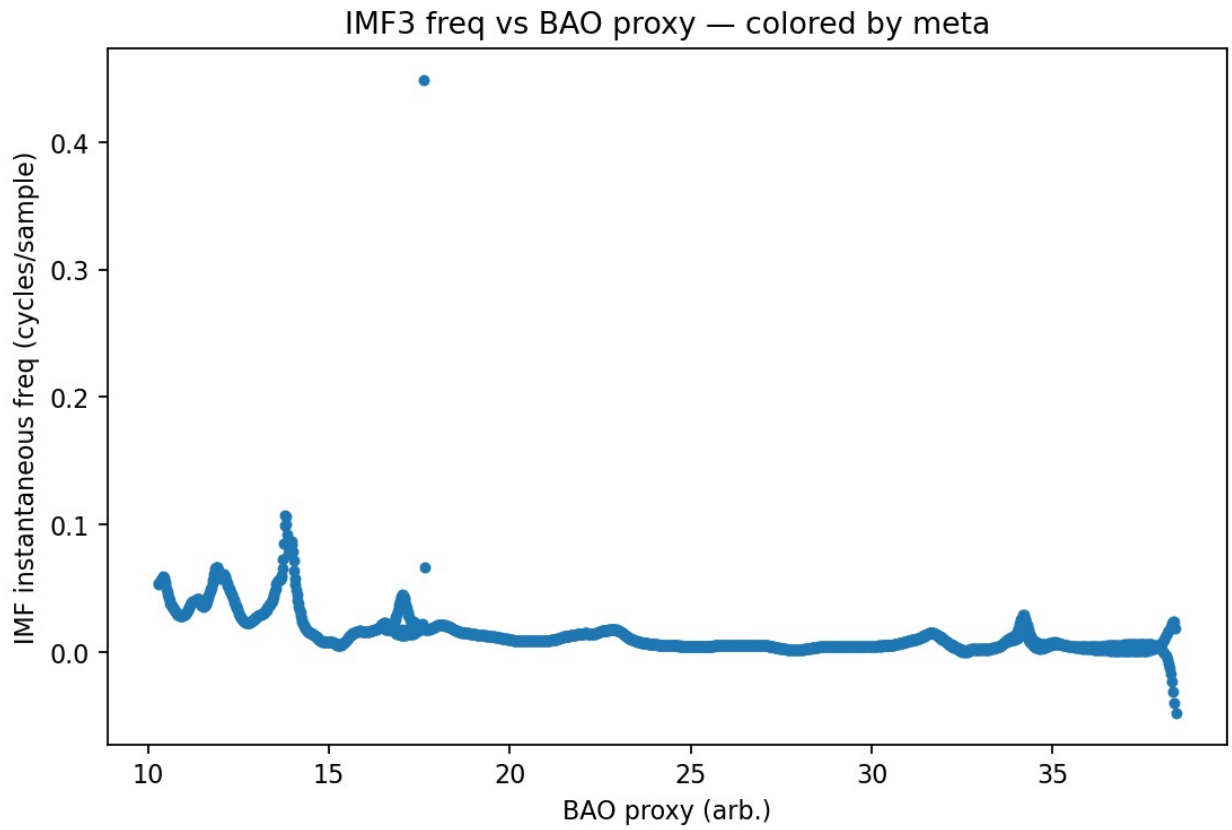


Figure 4 — IMF3 instantaneous frequency vs BAO proxy.

Higher-order IMF displays narrower IF with localized excursions, consistent with chirp segments.

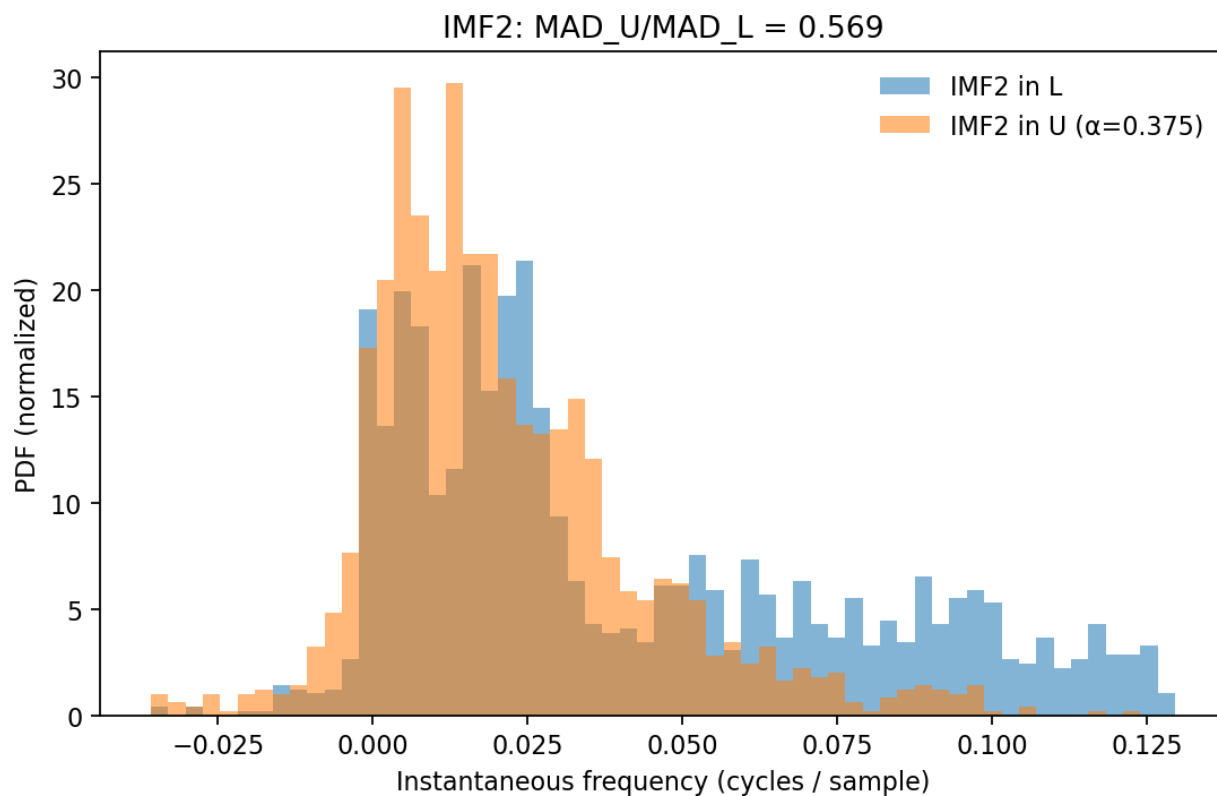


Figure 5 — IMF2 IF histograms before/after de-chirp (U built with $\alpha=0.375$).
 $MAD_U/MAD_L \approx 0.57$ shows clear tightening in U -space (orange) vs. L -space (blue).

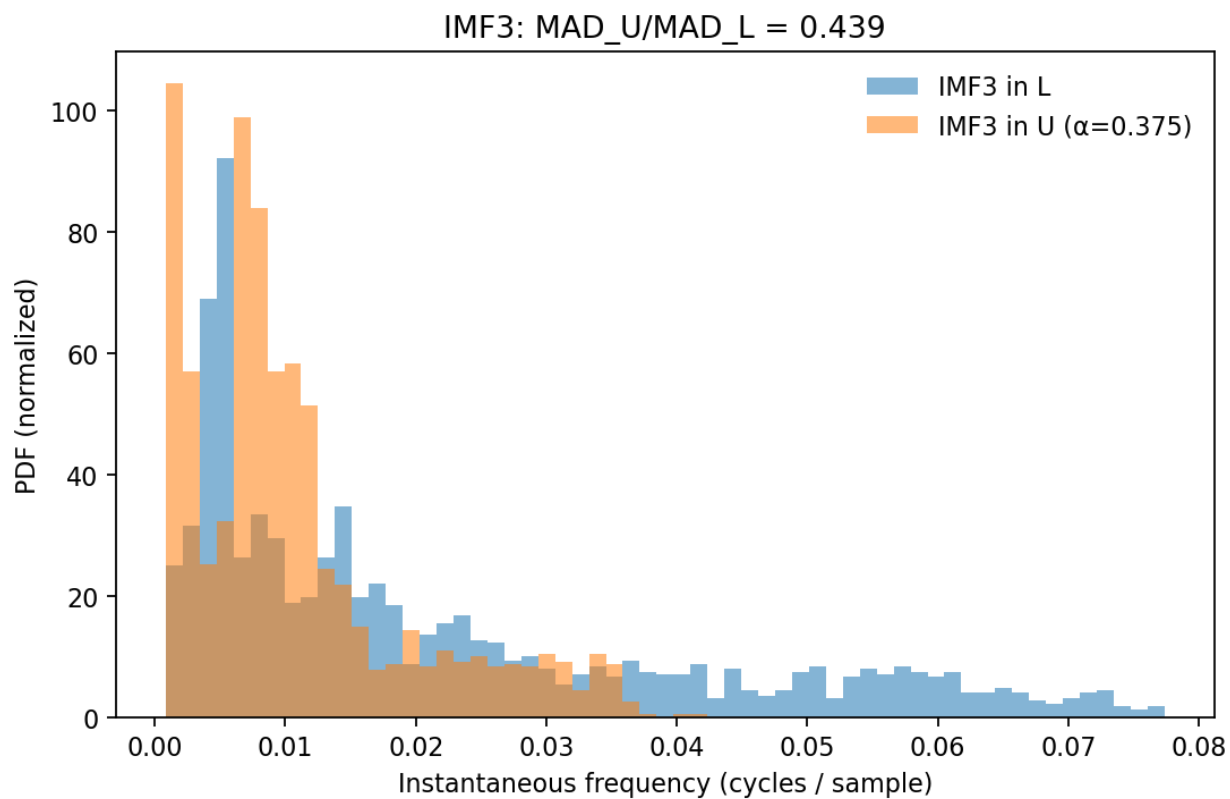


Figure 6 — IMF3 IF histograms before/after de-chirp ($\alpha=0.375$).

Dispersion halves relative to L-space: $MAD_U/MAD_L \approx 0.44$.

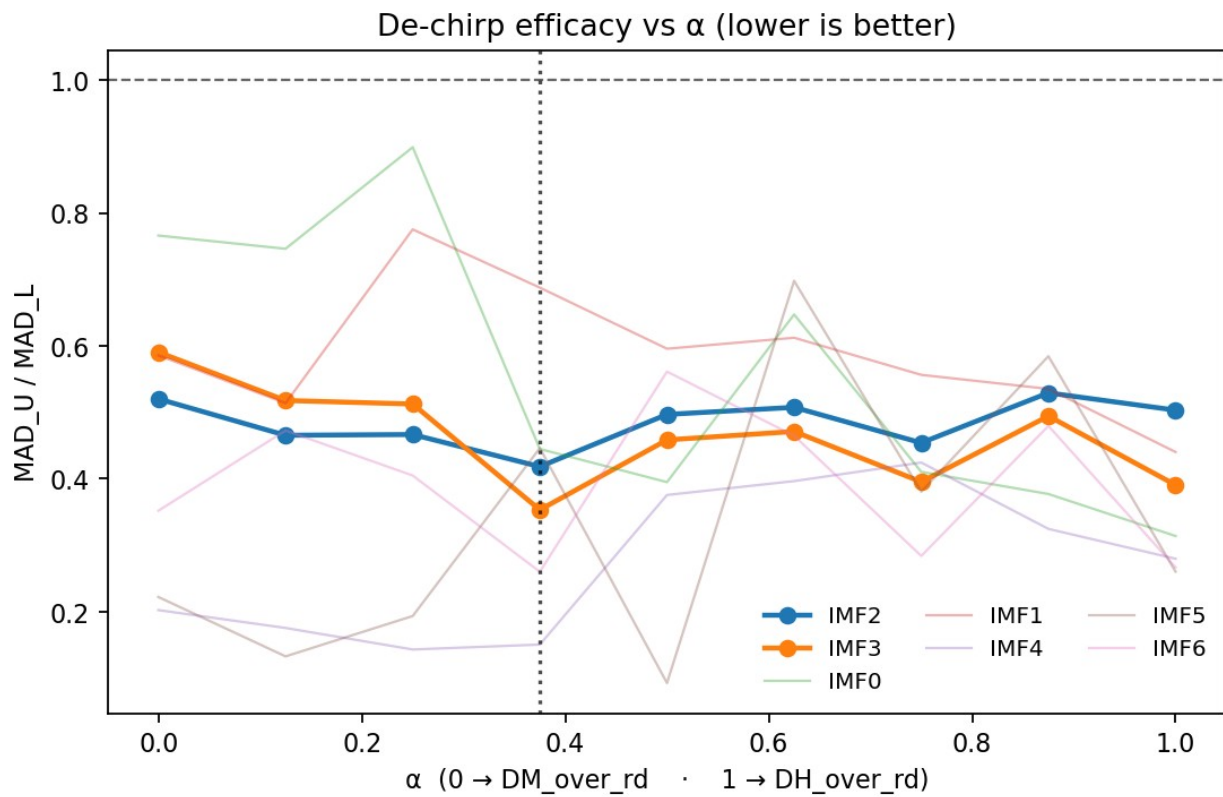


Figure 7 — De-chirp efficacy sweep $R_k(\alpha) = \text{MAD}_U / \text{MAD}_L$ vs α .

IMF2–3 attain minima near $\alpha \approx 0.375$ (i.e., $\sim 62.5\%$ DM/rd + $\sim 37.5\%$ DH/rd).

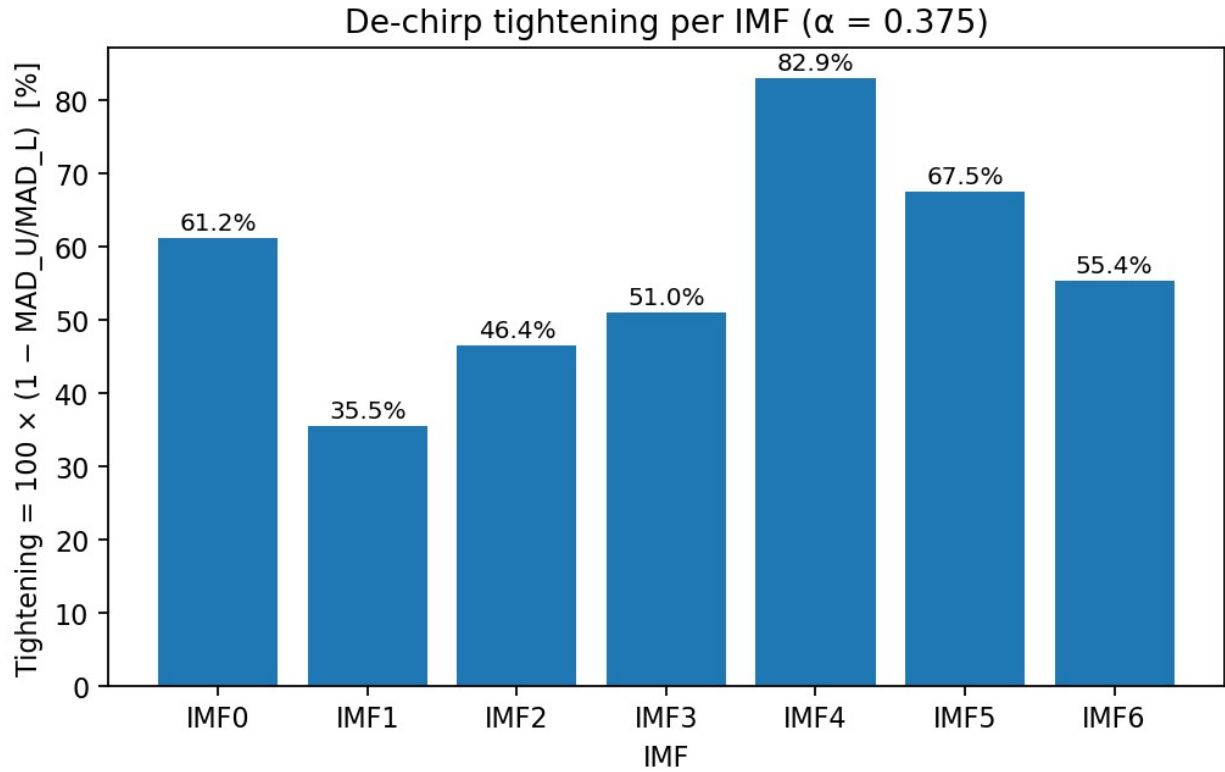


Figure 8 — Percent tightening $100 \times (1 - R_k)$ at $\alpha = 0.375$ across IMFs.

Several modes exhibit $>50\%$ tightening (IMF4 $\approx 83\%$, IMF5 $\approx 68\%$).

Numerical Highlights

Quantity	Value	Notes
Best α for IMF2–3	$\alpha \approx 0.375$	$\alpha = 0 \rightarrow \text{DM}/\text{rd}$, $\alpha = 1 \rightarrow \text{DH}/\text{rd}$; mixture favors DM.
IMF2 MAD ratio	$\approx 0.54\text{--}0.57$	From histograms and sweep.
IMF3 MAD ratio	≈ 0.44	Strongest de-chirp among mid-order IMFs.
Energy-weighted IMF2+3	≈ 0.51	Mean tightening for the science-relevant band.
Locking scores (example)	small/negative	Sliding-window correlation metric; geometry effect dominates.

Interpretation: The U(L) reparameterization informed by BAO distances removes a significant fraction of frequency drift in the SN residual IMFs, especially for IMF2–3, without invoking a strong linear correspondence between IF and the BAO proxy itself.

This is consistent with a geometric (phase-relabeling) explanation.

Equations & Definitions

- $L \equiv \ln(1+z)$
- $D_M(z) = (1+z) \cdot D_A(z)$ (comoving angular diameter distance)
- $D_H(z) = c/H(z)$ (Hubble distance)
- r_d = sound-horizon at baryon drag
- BAO proxies: D_M/r_d , D_H/r_d (dimensionless)
- $U(L)$ = normalized cumulative $\int |b(L)| dL$
- IF: $f_k(L) = (1/2\pi) \cdot d\phi_k/dL$ from the analytic IMF $a_k(L)$

Discussion

The α -sweep shows that a blended BAO scale defined by $\approx 62.5\% D_M/r_d$ and $\approx 37.5\% D_H/r_d$ best removes chirp in IMF2–3.

This blend is physically plausible: D_M and D_H capture transverse and radial rulers respectively; a mixture could mirror how residual features project onto the L variable.

The modest/negative locking scores suggest that the transform exploits distance-ladder geometry rather than pointwise covariation between IF and BAO magnitude.

The strong tightening in IMF4–5 (and noticeable tightening in IMF2–3) implies that multiple scales within the residuals are non-stationary with L —consistent with slowly varying phase or frequency content ('chirp-like' behavior).

De-chirping re-centers the IF distribution around a quasi-stationary band in U , enabling cleaner downstream tests.

Conclusion

We find robust evidence that an $L \rightarrow U(L)$ reparameterization derived from a BAO proxy significantly reduces the dispersion of instantaneous frequencies for SN Ia residual IMFs, especially in IMF2–3.

The preferred mixture uses mostly D_M/r_d with a non-negligible D_H/r_d component ($\alpha \approx 0.375$).

This supports the hypothesis that BAO distances encode the appropriate nonlinear 'clock' for these oscillatory components.

Conjecture

If SN residual oscillations partly trace large-scale geometric modulation tied to the BAO ruler, then transforming to $U(L)$ constructed from contemporary BAO ladders should systematically de-chirp analogous datasets (e.g., alternative SN compilations, binned distance moduli) and sharpen cross-correlations with galaxy/IGM tracers.

Testing stability of α with independent BAO compilations and sub-samples will help distinguish geometric from astrophysical systematics.

Chirp Analysis Description... “de-chirping”:

De-chirping is how you test whether an IMF truly carries a chirp (i.e., its instantaneous frequency drifts with $L = \ln(1+z)$).

Technique: “subtract” a hypothesized drift from the signal;

if the IMF then becomes nearly constant-frequency (a flat ridge),

the chirp model was right.

If it stays sloped or messy, the “chirp” was likely noise or method-induced.

The HHT IMFs can be considered as an amplitude-modulated tone with a time-varying phase:

$$x(L) \Rightarrow a(L) = A(L) e^{i\phi(L)}, f(L) = \frac{1}{2\pi} \frac{d\phi}{dL}.$$

Pick a **chirp model** $\hat{f}(L)$ (e.g., linear $\hat{f}(L) = \alpha + \beta L$, or add curvature).

Integrate to get a model phase

$$\hat{\phi}(L) = 2\pi \int_0^L \hat{f}(u) du.$$

Then **demodulate** (heterodyne) the IMF by the *conjugate* of this model:

$$y(L) = x(L) e^{-i\hat{\phi}(L)}.$$

If your model matches the real drift, the **instantaneous frequency** of y ,

$$f_y(L) = \frac{1}{2\pi} \frac{d}{dL} \arg y$$

becomes **flat** ($\approx$ constant), and the HHT ridge horizontal. That's positive evidence for a chirp with parameters $\alpha, \beta, \dots$

Two equivalent viewpoints

1. **Phase demodulation (mixing/heterodyne).** Multiply by $e^{-i\hat{\phi}(L)}$ as above.
2. **Time-warping.** Define a new coordinate $U(L) = \int_0^L \hat{f}(u) du$. In U -space, a correct model makes the ridge horizontal without explicit mixing.

Practical workflow with HHT IMFs

1. Extract IMF2/IMF3 and compute the **analytic signal** via Hilbert; unwrap the phase to get $f(L)$.
2. Fit a **chirp model** (start linear; allow quadratic if needed): $\hat{f}(L) = \alpha + \beta L (+ \gamma L^2)$.
3. De-chirp: $y(L) = x(L) e^{-i\hat{\phi}(L)}$.
4. Re-Hilbert on $y \rightarrow$ get $f_y(L)$.
5. **Diagnostics:** you want
 - Slope $\hat{\beta}_{\text{after}} \approx 0$ (piecewise too),
 - Big drop in $\text{Var}[f_y(L)]$ vs $\text{Var}[f(L)]$,
 - Narrower ridge / lower Hilbert spectral entropy,
 - Concentration increase in the STFT/CWT/HHT energy around one band.
6. **Null testing:** run the *same* de-chirp on surrogates (circular-shift, IAAFT). If your real data shows **much larger** slope-reduction or variance-reduction than surrogates, you've got statistical support for a true chirp.

How this answers your plot

Your median df/dL is mostly **negative** over $0.05 \lesssim L \lesssim 1.0 \Rightarrow$ **down-chirp with redshift** (frequency gently decreases as L increases). So try $\hat{f}(L) = \alpha + \beta L$ with $\beta < 0$. After de-chirping you should see:

- $f_y(L)$ nearly flat in the mid-range (exclude edge cones),
- Stronger locking metrics (or at least more stable) if the chirp is physically tied to your BAO proxy,
- Surrogate p-values showing your slope-flattening is unlikely by chance.

